Graph Sketching: Reciprocal Hyperbolic Functions

Questions 1–8

Question 1. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

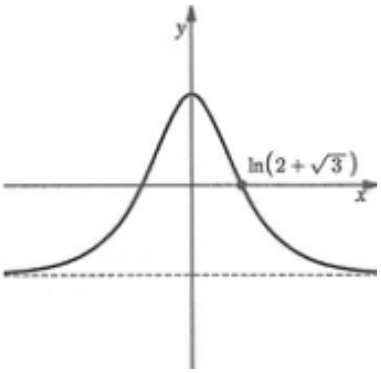
a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

Question 2.

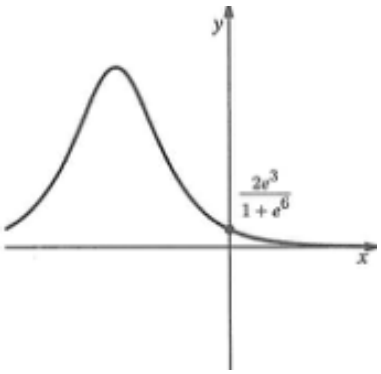
a) By sketching the curves with equations $y = \cosh x$ and $y = \operatorname{cosech} x$ on the same axes, show that they have exactly one point of intersection.
b) Find the x-coordinate of the point of intersection of the two curves, writing your answer in the form $\frac{1}{2}\ln(a+\sqrt{b})$, where a and b are positive integers.

Question 3. Find the equation of each of the following graphs in the form $y = \text{sech}(x - a)$ or $y = \text{sech}\, x + b$, where a and b are integers:

a)



b)



Question 4. State the equations of the asymptotes of the following graphs:

a) $y = \coth(x + 1) + 1$

b) $y = \coth(x - 2) - 3$

Question 5. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \coth x + 2$
b) $y = \coth(x - 3)$
c) $y = \coth(x - 1) + 1$

Question 6. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 2\coth x$
b) $y = \frac{1}{3}\coth x$
c) $y = -\coth x$

Prerequisite knowledge for Q7–Q8: Combinations of Graph Transformations

Question 7. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 2 - \operatorname{cosech} x$
b) $y = -\frac{1}{3}\operatorname{sech} x$
c) $y = 2\coth x - 1$
d) $y = \operatorname{cosech}(2x + 1)$
e) $y = \operatorname{sech}(3 - x)$
f) $y = \coth\left(\frac{1}{2}x - 3\right)$

Question 8. Suggest an equation for the curve shown in the diagram below:

